

Loop ■ — Technical Architecture & Specification

Autonomous Multi-Agent Conversational Language Learning System with Spaced-Repetition Memory & Dynamic Error Retriggering

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Track: Language Learning App	Code Repo: github.com/shubhampatil631/Loop
Demo Video: youtu.be/0D93_SNgMBI	Stack: FastAPI, LangGraph, ChromaDB, Mongo, React+TS, MCP

1. Executive Summary & Problem Solved

Current digital language learning tools suffer from a clear dichotomy: static flashcard apps (e.g. Anki, Duolingo) rely on rote memorization without situational fluency, while open-ended LLM chatbots lack pedagogical memory, fail to track learner mistakes systematically, and hallucinate vocabulary outside the user's comprehension level.

Loop bridges this gap by delivering an end-to-end, multi-agent educational pipeline. It conducts natural conversational roleplay strictly grounded in CEFR-appropriate vocabulary, silently classifies mistakes without interrupting dialogue flow, tracks individual memory decay via the FSRS algorithm, and dynamically re-injects past errors and overdue words into subsequent scenarios until mastery is proven.

2. Multi-Agent Orchestration Engine (LangGraph)

Loop's cognitive architecture is implemented as a stateful, compiled LangGraph graph comprising six specialized, modular agents communicating over a unified state machine:

Agent Name	Core Responsibility	Technical Implementation & Role
Placement Agent	CEFR Diagnostic Baseline	Runs an adaptive 3-4 turn diagnostic interview to assess grammatical range, lexical breadth, and coherence to assign A1–B2 baseline levels.
Conversation Partner	Immersive Grounded Roleplay	Simulates rich real-world scenarios (boulangerie, airport, cafe). Dynamically queries ChromaDB RAG to constrain language strictly within CEFR ceiling.
Error-Analysis Agent	Non-blocking Diagnostics	Asynchronously parses learner inputs to detect and tag mistake taxonomy (conjugations, gender agreements, false friends, syntax) with corrections.
Curriculum Scheduler	Spaced Repetition (FSRS)	Computes stability, ease factors (EF), and review intervals. Dynamically re-queues lapsed vocabulary and past mistake tags for future scenario insertion.
Review Recall Agent	Active Recall Verification	Administers pre-session targeted flashcard drills, calculating active recall response accuracy and updating memory retention scores.
Weekly Digest Agent	Longitudinal Mastery Analytics	Aggregates historical session data into mastery percentages, retention curves, mistake heatmaps, and prescriptive weekly improvement plans.

3. Key Technical Innovations

A. Resilient 3-Tier LLM Cascading Fallback Engine

To achieve zero-downtime reliability against API rate limits and network outages, Loop implements an automatic 3-tier cascade: **Tier 1: Google Gemini 1.5 Flash** (Cloud Primary) → **Tier 2: Groq LLaMA 3.3 70B Versatile** (Ultra-fast Cloud Backup) → **Tier 3: Local Ollama (Qwen 2.5 / LLaMA 3.2)** (Air-gapped on-device fallback).

B. FSRS Spaced-Repetition Memory Matrix & Dynamic Error Retriggering

Unlike standard flashcards, Loop calculates dynamic memory decay curves using Free Spaced Repetition Scheduler (FSRS) formulas. When a user fails a grammatical concept or vocabulary item, the Curriculum Agent silently flags it as an active trigger. In future roleplay sessions, the Conversation Partner naturally engineers situational prompts forcing the learner to apply that specific concept again.

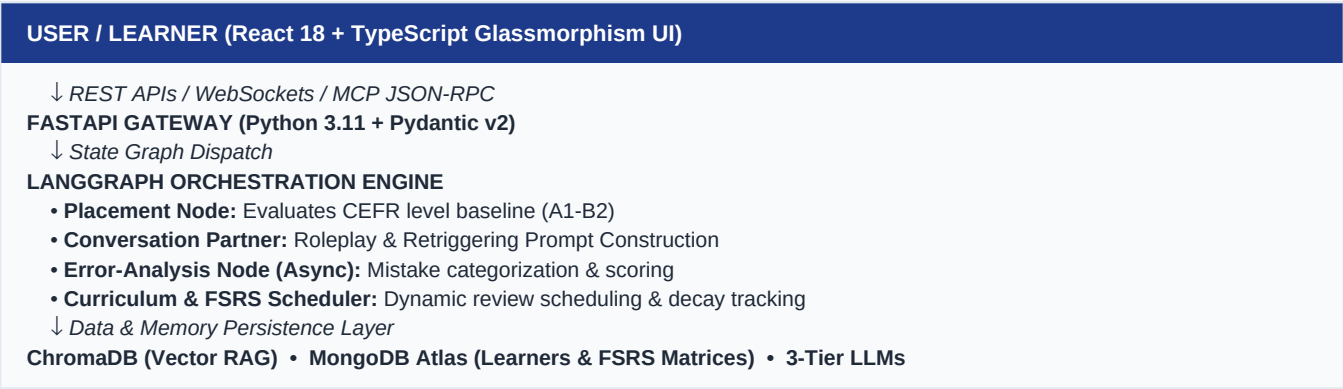
C. RAG-Grounded Vector Vocabulary (ChromaDB)

Hallucination prevention is enforced by vectorizing 200+ curated CEFR vocabulary targets using all-MiniLM-L6-v2 embeddings. Queries are filtered by CEFR hierarchy ($A1 \subseteq A2 \subseteq B1 \subseteq B2$) and contextual scenario tags.

D. Standardized Model Context Protocol (MCP) Server

Loop features a native MCP server exposing JSON-RPC 2.0 endpoints (get_due_items, fetch_level_vocab, log_error_tag, schedule_next_review, get_learner_profile) enabling external agent interoperability.

4. End-to-End System Flow & Architecture Diagram



5. Full Technology Stack & Deliverables

Layer	Technologies & Frameworks
Backend	Python 3.11, FastAPI, Uvicorn, LangGraph, LangChain, Pydantic v2
AI Models & Fallback	Google Gemini 1.5 Flash (Tier 1), Groq LLaMA 3.3 70B (Tier 2), Ollama Qwen/LLaMA (Tier 3)
Vector Store & Embeddings	ChromaDB persistent vector database with all-MiniLM-L6-v2 embeddings
Database	MongoDB Atlas & Local MongoDB instance via PyMongo
Frontend UI	React 18, TypeScript, Vite, Lucide Icons, Custom Glassmorphic Dark UI
Protocol & Integration	Model Context Protocol (MCP) Server & Standard JSON-RPC 2.0 Tools

DevOps & Packaging	Docker, Docker Compose, Pytest automation suites
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6. Future Roadmap

- **Full-Duplex Voice & Pronunciation:** Streaming WebRTC speech-to-text with phoneme scoring.
- **Multimodal Contextual Stimuli:** Visual roleplay elements (menus, street signs, tickets).
- **Multi-Language Expansion:** Grammar taxonomies for Japanese, German, Mandarin, and Spanish.
- **Edge / Mobile Deployment:** React Native client with local SLM execution on Apple Silicon / Snapdragon.